NeST-Com: Fixed-Budget Next-Scale Token Communication

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Abstract

Next-scale residual tokens provide a coarse-to-fine image description, but whole-scale endpoints can underfill fixed digital frames. NeST-Com fits independently terminated, entropy-coded within-scale prefixes to fixed frames and uses received tokens for residual reconstruction and finer-scale conditioning. With shared model components, it improves PSNR over whole-scale transmission by 0.32–0.70 dB at 1024 symbols and 0.39–0.79 dB at 512 symbols. At 1024 symbols and 10 dB, source-bit utilization increases from 77.70% to 95.21%. Comparisons with existing systems show favorable low-budget source reconstruction and metric-dependent perceptual tradeoffs.

Index Terms

Image communication, visual autoregressive models, fixed symbol budget, entropy coding.

I. Introduction

A fixed per-image channel-symbol budget limits the source description available for reconstruction. Its useful content depends on image-dependent source-code length and the protection carried in the same frame. A generative receiver can complete untransmitted information, while the received description constrains that completion to the source image. The challenge is to retain as much useful source information as the frame permits and to use it throughout reconstruction.

Next-scale representations organize an image into complete coarse token maps followed by residual refinements. After resizing, each coarse map covers the full latent grid, providing a global description before finer maps add detail. Visual autoregressive modeling (VAR) reported state-of-the-art results in its original class-conditional ImageNet generation benchmarks [7], motivating its use as a completion prior. Digital coding and learned joint source–channel coding (JSCC) address limited channel resources [1], [2]; adapting progressive descriptions is a classical problem [3]. Recent token schemes include Ada-TokenCom's prefix coding and tail generation [4], task-oriented TONIC [5], and next-scale progressive ARPC [6].

Whole-scale boundaries need not align with fixed frame capacity: a scale adds many tokens, and the terminated entropy-code length varies across images. Legal endpoints inside a scale can retain source tokens that would otherwise be excluded. VAR's coarse-conditioned factorization lets both terminals construct matching probabilities for these positions from complete coarse maps. The delivered boundary tokens must then refine both residual accumulation and the context for predicting finer scales.

NeST-Com combines within-scale endpoints, independently terminated arithmetic streams, actual-length fitting, and prefix-constrained reconstruction. Using VAR's known-token overwrite [7], it merges received positions with predictions and updates both the residual latent and finer-scale context. With shared model components, it improves PSNR over whole-scale transmission by 0.32–0.70 dB at 1024 symbols and 0.39–0.79 dB at 512 symbols. At 1024 symbols and 10 dB, source-bit utilization rises from 77.70% to 95.21%. Controlled experiments examine the endpoint and receiver designs, while complete-system comparisons show their reconstruction tradeoffs.

II. Proposed NeST-Com Framework

A. Next-Scale Representation and Digital Frame

We transmit $x \in [0,1]^{256\times256\times3}$ over complex AWGN,

$$y = s + n, \quad n \sim \mathcal{CN}(0, \sigma^2 I_N), \quad \mathrm{SNR} = 10\log_{10}(E_s/\sigma^2), \tag{1}$$

using N complex symbols per image and nominal $E_s = 2$. TX selects a frozen policy for the channel SNR; RX uses the true noise variance for LLRs. Digital constellations have average energy two without per-frame renormalization. Continuous JSCC data are normalized per image to average energy two, with required Swin side information paid within N. A 12-bit profile ID, CRC16, and terminated rate-1/2 convolutional code occupy 68 rotated-BPSK symbols $\{\pm(1+j)\}$. The remaining $N-68$ symbols carry the LDPC-coded body. After rate matching, a public frame-dependent scrambling mask is applied before QAM modulation and reversed at the receiver before LDPC decoding. The paid profile identifies the actual endpoint and MCS. Our main budget is $N = 1024$; the reduced-budget experiment uses $N = 512$ without increasing E_s.

The frozen VAR tokenizer produces ten residual token maps q_s, with side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$ and a 4096-entry codebook [7]. Let h_s denote the side length, $L_s = h_s^2$ the number of positions, and C_s the scale-dependent codebook lookup, resizing, and residual transform. Complete coarse maps span the whole latent grid after resizing; the sum $f_q = \sum_s C_s(q_s)$ accumulates their residual refinements. Image decoder D_c is adapted on 20 000 ImageNet training images using continuous, fully quantized, and prefix-completion-interpolated latents, then fixed. NeST-Com uses null-class conditioning. Its frozen VAR-d16 next-scale prior factorizes [7] as

$$p_\theta(q_s \mid q_{<s}) = \prod_{i=1}^{L_s} p_\theta(q_{s,i} \mid q_{<s}), \tag{2}$$

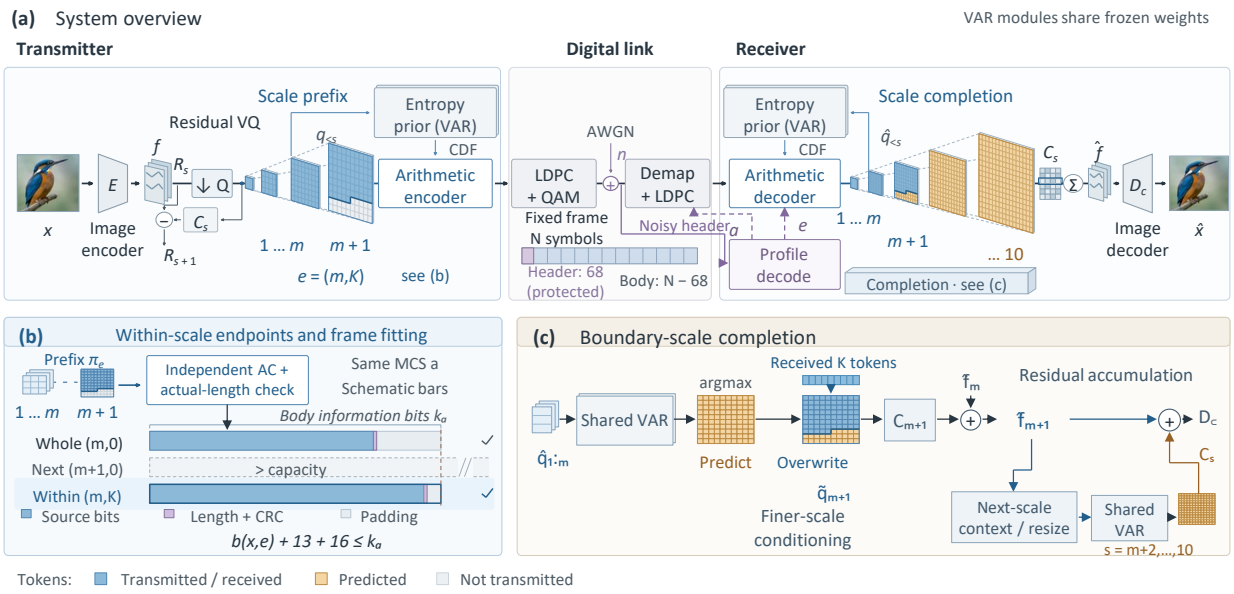


Fig. 1. NeST-Com: (a) The protected profile and body share the fixed noisy frame; both terminals reconstruct coding probabilities from the shared frozen VAR and known coarse tokens. (b) Independently terminated within-scale prefixes fit the body information capacity; bars are schematic for the same MCS. (c) Received tokens overwrite boundary predictions, and the merged map updates the residual latent and finer-scale context while unreceived positions retain their coarse-conditioned predictions.

All positions in a scale thus obtain coding probabilities from the same complete coarse maps. Both terminals can construct identical cumulative distribution functions (CDFs) for arithmetic coding [8] of a transmitted within-scale subset (Fig. 1).

B. Within-Scale Endpoints and Actual-Length Fitting

An endpoint $e = (m, K)$ retains complete coarse maps and K raster-ordered positions from the next map:

$$\pi_e = (q_1, \ldots, q_m, q_{m+1,1:K}). \tag{3}$$

For $K = 0$, the boundary tail is absent. WHOLE permits $K = 0$ for $m = 4, \ldots, 10$. PARTIAL adds $K \in \{\lfloor L_{m+1}/4 \rfloor, \lfloor L_{m+1}/2 \rfloor, \lfloor 3L_{m+1}/4 \rfloor\}$ for $m = 4, \ldots, 9$, giving 25 endpoints in total. $(10, 0)$ is the full representation.

For MCS $a = (M, r)$, the coded body has $n_a = (N - 68) \log_2 M$ bits and information length $k_a = \lfloor r n_a \rfloor$. With actual terminated arithmetic length $b(x, e)$,

$$b(x, e) + 29 \leq k_a, \qquad \eta(x, e) = \frac{b(x, e)}{k_a - 29}. \tag{4}$$

The endpoint specifies retained source tokens; its actual terminated length determines frame feasibility. The 29 bits comprise the 13-bit length field and CRC16; unused information positions are zero-padded before 5G NR LDPC coding, rate matching [9], and QAM. Thus η measures source-bit capacity utilization, not total over-the-air efficiency.

Offline calibration selects a target endpoint and MCS separately for each budget and SNR on an independent image set. It maximizes source/reconstruction DINOv2 cosine [10], averaged first over channel realizations and then equally over sources. The selected policy is frozen for testing; candidate pools and screening rules are given in the accompanying configuration material.

Online fitting fixes the selected MCS and searches legal endpoints no longer than the target, in decreasing token count down to $(4, 0)$. TX sends the first independently terminated stream satisfying (4); the paid profile identifies the actual endpoint. Each candidate prefix is independently arithmetic-coded and terminated, rather than cut from a longer stream. This source-coding operation leaves the body-level acceptance rule unchanged. If no endpoint fits, the N512 protocol sends a paid NULL profile and short NULL payload through the complete frame, retaining a gray reconstruction in the denominator. The evaluated N1024 implementation instead raises an error; no evaluated N1024 source encountered this case.

C. Prefix-Constrained Reconstruction

Invalid profile CRC/catalogue, body CRC, declared length, known padding, or arithmetic parsing produces a fixed gray image; so does a valid N512 NULL reception. Otherwise, RX decodes the prefix using CDFs rebuilt from received complete coarse maps. It predicts the boundary map by VAR argmax conditioned on those maps, then applies known-token overwrite [7]:

$$\tilde{q}_{m+1,i} = \begin{cases} \hat{q}_{m+1,i}, & i \le K, \\ q^{\mathrm{pred}}_{m+1,i}, & i > K. \end{cases} \tag{5}$$

Unreceived boundary positions retain their coarse-conditioned prediction. Received boundary tokens serve two roles: they correct the residual representation and, through the merged boundary map, condition the prediction of finer scales. Complete received maps remain unchanged; subsequent maps are generated from this updated context. Reconstruction is

$$\hat{x} = D_c\left(\sum_{s=1}^{10} C_s(\tilde{q}_s)\right). \tag{6}$$

At the final scale, overwrite changes the residual contribution without any subsequent propagation.

III. Experimental Results

A. Experimental Setup

We use common500 ImageNet validation images [11], a disjoint cal1000 set, and Kodak24 [12], with fixed 256×256 preprocessing. ImageNet-calibrated policies remain fixed on Kodak. The six-system main study uses $N = 1024$; NeST, WHOLE, and Ada also use $N = 512$, each independently calibrated at 4/10/19 dB.

Our convolutional Latent JSCC maps tokenizer latents to symbols and shares D_c, using MSE, LPIPS, and latent-consistency losses. It and SwinJSCC [2], adapted to 256×256 and

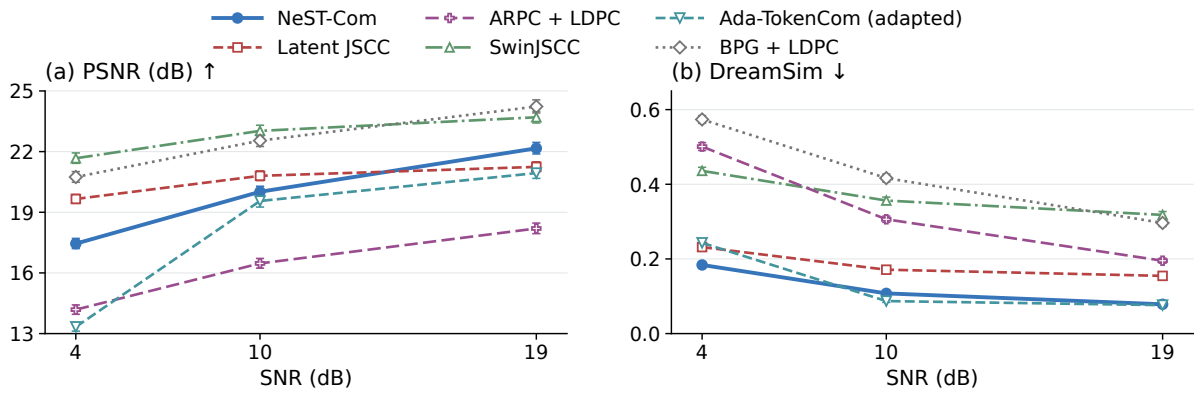


Fig. 2. Performance on 500 ImageNet validation images at $N = 1024$. Error bars show 95% source-bootstrap confidence intervals. Ada denotes the fixed-frame adaptation.

N=1024, use the same 20 000 training images and final 100k/200k-update checkpoints. Digital baselines are adaptive BPG [13] + LDPC, ARPC [6] + LDPC with paid captions, and Ada-TokenCom [4] adapted to the same fixed single-frame budget. Ada retains LlamaGen-L/VQ16 and prefix arithmetic coding, with a fixed ConvNeXt predicting a paid 10-bit class. Its sampled tail uses CFG=4, $T = 1$, top-$k = 2000$, and top-$p = 1$. Full configurations and training records accompany this letter.

Metrics are PSNR, LPIPS-Alex [14], DreamSim [15], and CLIP ViT-L/14 image–image cosine (CLIP-I) [16]. Frozen FP32 preprocessing uses 224×224 bicubic-antialias inputs for DreamSim/CLIP. Three channel outcomes per image, including the gray outputs defined in Sec. II-C, are averaged before equal-source means and direct source-paired 95% bootstrap intervals (10 000 resamples). Endpoint, receiver, and decoding controls use a fixed 100-image subset. Complete four-metric diagnostics and statistical identities are provided in the supplementary material.

TABLE I

WITHIN-SCALE ENDPOINT AND RECEIVER EVIDENCE. (A) MAIN FAMILY-SPECIFIC POLICIES, PARTIAL MINUS WHOLE. (B) FIXED 100-SOURCE MODEL/MCS/MAXIMUM-ENDPOINT CONTROL. (C) ORDERED INTERVENTIONS ON THE SAME 100 SOURCES AND ACTUAL PREFIXES. BRACKETS GIVE SOURCE-PAIRED 95% CIS.

(A) Main policies at $N = 1024$

SNR	Util. W→P (%)	ΔPSNR (dB) [95% CI]
common500		
4	73.81 → 93.08	$+0.616$ $[0.564, 0.673]$
10	77.70 → 95.21	$+0.701$ $[0.637, 0.766]$
19	85.24 → 94.73	$+0.319$ $[0.290, 0.349]$
Kodak24		
4	75.69 → 93.43	$+0.396$ $[0.236, 0.543]$
10	78.11 → 94.99	$+0.674$ $[0.408, 0.958]$
19	85.69 → 93.71	$+0.228$ $[0.140, 0.324]$

(B) Fixed-granularity diagnostic, 10 dB

Grid (candidates)	Utilization (%)	PSNR (dB)
Whole (7)	78.15	18.952
Half (13)	89.21	19.396
Quarter (25)	95.38	19.612
Eighth (49)	97.39	19.735

(C) Receiver interventions: ΔPSNR (dB) [95% CI]

SNR	B0−A0	C0−B0
10	$+0.220$ $[0.176, 0.264]$	$+0.440$ $[0.330, 0.558]$
19	$+0.301$ $[0.247, 0.358]$	$+0.000$ $[0.000, 0.000]$

Boundary tokens: A0, unused; B0, residual correction only; C0, residual correction and conditioning update. All three use the received complete coarse scales.

B. Effect of Within-Scale Endpoints

Table I(A) compares WHOLE/PARTIAL with shared tokenizer, prior, entropy coder, and D_c. Separate family calibration selects the same MCS at each SNR. Common500 PSNR gains are 0.616/0.701/0.319 dB at 4/10/19 dB. At 10 dB, source-bit utilization rises from 77.70% to 95.21% and DreamSim falls by 0.0240. On Kodak, the corresponding PSNR gain is 0.674 dB and DreamSim decreases by 0.0164; its 19 dB DreamSim interval includes zero. The higher utilization retains additional source tokens within the same frame.

Block B fixes the 100 sources, model, MCS, maximum endpoint, and fitting rule. Whole/Half/Quarter/Eighth grids increase utilization from 78.15% through 89.21% and 95.38% to 97.39%, with PSNR increasing from 18.952 to 19.735 dB. Relative to the predefined Quarter configuration, Eighth adds $0.123\,[0.079, 0.169]$ dB at 10 dB; its DreamSim/CLIP-I difference intervals cross zero.

Block C applies the boundary-token interventions defined in the table to the same 100 sources and actual received prefixes. The ordered PSNR gains are 0.220 and 0.440 dB at 10 dB, supporting residual correction and the subsequent conditioning update. At 19 dB, the boundary lies in the last scale, so the additional conditioning update has zero effect.

Within the fixed adapted receiver, completion improves the three perceptual measures over coarse-only decoding at both tested low-budget points; complete results are provided in Supplement S4.

C. Comparison With Existing Methods

Fig. 2 and Table II show the complete-system tradeoffs at $N = 1024$. NeST exceeds Ada in PSNR by 4.116/0.455/1.226 dB at 4/10/19 dB and exceeds ARPC by 3.26–3.97 dB. Ada has better LPIPS, DreamSim, and CLIP-I at 10 dB, higher CLIP-I at 4 dB, and slightly lower

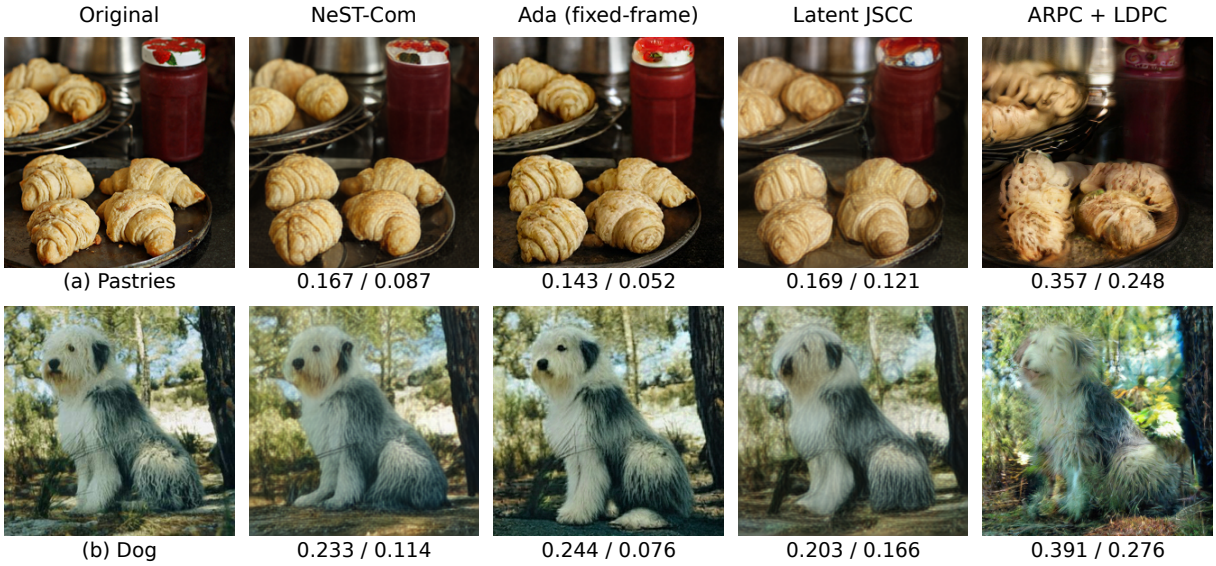


Fig. 3. Qualitative reconstructions of two development examples at $N = 1024$ and 10 dB. Values denote LPIPS-Alex↓ / DreamSim↓. Ada denotes the fixed-frame adaptation.

TABLE II

PERCEPTUAL DISTANCE AND IMAGE-FEATURE SIMILARITY ON COMMON500 AT $N = 1024$. EACH CELL IS LPIPS-ALEX↓ / CLIP-I↑. ADA DENOTES THE FIXED-FRAME ADAPTATION. VALUES ARE SOURCE MEANS.

Method	4 dB	10 dB	19 dB
NeST-Com	0.263 / 0.821	0.171 / 0.873	0.118 / 0.900
Latent JSCC	0.219 / 0.741	0.173 / 0.789	0.158 / 0.804
SwinJSCC	0.530 / 0.627	0.469 / 0.669	0.440 / 0.697
BPG + LDPC	0.725 / 0.586	0.589 / 0.635	0.458 / 0.684
ARPC + LDPC	0.512 / 0.644	0.341 / 0.713	0.250 / 0.788
Ada (adapted)	0.395 / 0.828	0.167 / 0.884	0.138 / 0.892

DreamSim at 19 dB. SwinJSCC and BPG have higher PSNR than NeST throughout; Latent JSCC has higher PSNR at 4/10 dB and lower LPIPS at 4 dB. Thus the source-constrained

generative route offers a different balance between pixel fidelity and perceptual reconstruction. In the two development examples of Fig. 3, Latent JSCC smooths pastry ridges and facial details, while ARPC changes pastry shapes and the dog's pose. NeST and Ada preserve more of these structures but differ in detail; the displayed metric rankings vary by image.

D. Transmission With a Reduced Symbol Budget

TABLE III

REDUCED-BUDGET COMPARISON ON COMMON500 AT $N = 512$. DIFFERENCES ARE NEST-COM MINUS THE FIXED-FRAME ADA-TOKENCOM ADAPTATION; BRACKETS GIVE SOURCE-PAIRED 95% CIS. POSITIVE PSNR AND NEGATIVE DREAMSIM FAVOR NEST-COM.

SNR	ΔPSNR (dB)	ΔDreamSim
4	$+4.180$	-0.1029
	$[3.991, 4.377]$	$[-0.1142, -0.0917]$
10	$+4.055$	-0.0660
	$[3.880, 4.235]$	$[-0.0747, -0.0573]$
19	$+1.740$	$+0.0139$
	$[1.577, 1.909]$	$[0.0103, 0.0173]$

At $N = 512$, 68 header symbols leave 444 body symbols. NeST improves over WHOLE by $0.392\,[0.341, 0.444]$, $0.399\,[0.356, 0.445]$, and $0.792\,[0.740, 0.844]$ dB at 4/10/19 dB. All corresponding LPIPS, DreamSim, and CLIP-I paired intervals also favor NeST. At 10 dB, WHOLE and NeST select the same target $(7, 0)$ and 16-QAM rate-2/3 MCS. Both fall back from this target for 99.6% of sources, but NeST retains a within-scale prefix for 89.0%. The mean number of retained tokens increases from 91.3 to 107.9, raising source-bit utilization from 79.30% to 93.46%.

WHOLE already has higher mean PSNR than Ada by 3.79/3.66/0.95 dB. NeST increases these gaps to 4.180/4.055/1.740 dB (Table III) and has lower LPIPS at all three SNRs. DreamSim favors NeST at 4/10 dB and Ada at 19 dB; CLIP-I favors Ada throughout.

Replacing Ada's sampled tail by argmax on the fixed 100-image subset leaves NeST higher in PSNR by 4.63/4.60 dB at $N = 512$, 4/10 dB; complete sensitivity results are in Supplement S3.

IV. Conclusion

NeST-Com adapts next-scale residual representations to fixed digital frames through within-scale endpoints, actual-length fitting, and prefix-constrained reconstruction. Received boundary tokens improve both residual reconstruction and finer-scale prediction. Shared-model experiments show higher source-bit utilization and consistent PSNR gains at 512- and 1024-symbol budgets, while comparisons with existing systems show useful low-budget reconstruction performance with metric-dependent perceptual tradeoffs. Future work will extend frame selection to uncertain channel conditions.

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